

Structo Robotics

Applied robotics for safer construction assembly and smarter solar-panel maintenance.



Two focused pre-PoC programs

● Robotic precast assembly

A retrofit automation layer for an existing high-reach heavy platform: four-point handling, computer vision, motion control and an auxiliary bolting manipulator.

● Solar inspection and cleaning robot

An operator-supervised robotic platform that detects panel soiling, maps priority areas and performs controlled lightweight cleaning with before-and-after data.

Team

Shlomo Eidelstein, founder and engineer, supported by a software developer and an architect. The team combines field installation experience, software delivery and practical understanding of solar and building systems.

STAGED FINANCING TARGET

Up to USD 1.5M

Phase 1 - USD 350k: engineering validation, bench prototypes, safety architecture and pilot agreements.

Phase 2 - USD 1.15M: field prototypes, pilot deployment, certification, IP work and commercial readiness for both programs.

Open to accelerator investment, non-dilutive funding, corporate co-development, in-kind engineering and pilot access.

Robotic assembly of precast concrete elements

A practical automation layer for existing heavy machinery, designed to reduce work-at-height exposure and improve positioning repeatability on modular construction sites.

Existing platform

Use a digitally controlled high-reach demolition excavator platform, such as a Volvo EC950-class machine, instead of developing heavy machinery from zero.

Robotics layer

Add four-point load handling, sensing, computer vision and closed-loop motion control to stabilize and place precast modules.

Connection step

Use a secondary manipulator to align prepared joints, install bolts, control tightening torque and record the result.

Initial proof-of-concept targets

Up to 3 t

Demonstration payload

≤ 2 cm

Positioning target

95%+

Repeatability target

10+

Placement cycles

COLLABORATION ASK

Within a combined staged financing target of up to USD 1.5M, we seek a construction-tech accelerator, heavy-equipment or precast partner, engineering mentor and controlled pilot site. Funding and in-kind support will be tied to a safe bench-scale POC, field validation and measurable milestones - not an autonomous commercial machine at the outset.

Solar-panel inspection and cleaning robot

A measured maintenance workflow for dusty rooftop and utility-scale photovoltaic assets: inspect, prioritize, clean locally and document the result.

● Measure first

Camera and sensor data estimate soiling, identify priority zones and create a repeatable before-and-after record.

● Controlled cleaning

A lightweight soft-cleaning mechanism is tested on a supervised robotic or aerial platform with emergency stop and safety procedures.

● Service pathway

Start with monitoring and cleaning services, then expand toward software subscriptions, fleet operations and technology licensing.

Initial proof-of-concept targets

12 mo

POC roadmap

85%+

Soiling detection target

90%+

Test-area cleaning target

1 pilot

Partner validation site

COLLABORATION ASK

Within a combined staged financing target of up to USD 1.5M, we seek a climate- or energy-tech accelerator, PV asset owner, O&M; partner, robotics mentor and test site. Funding and in-kind support will validate cleaning quality, safe operation, unit economics and the preferred platform architecture through a supervised field pilot.